

STUDENT RECORD MANAGEMENT SYSTEM

Using Singly Linked List in C Programming

Comprehensive Project Report

Prepared from analysis of headersll.h and sll.c source files.

Certificate

This report presents the design and implementation of a Student Record Management System using a Singly Linked List. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Abstract

The project demonstrates dynamic memory allocation, linked list operations, recursion, sorting, searching, deletion and file handling. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Introduction

A Singly Linked List is a dynamic data structure where each node stores data and the address of the next node. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Objectives

Create, store, search, sort, delete and manage student records efficiently. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Software Requirements

C Compiler, Linux/Windows environment, GCC, standard libraries stdio.h, stdlib.h, string.h and unistd.h. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Data Structure Design

Structure SLL contains rollno, name, marks and next pointer. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Program Flow

User selects an operation from a menu. The selected function manipulates the linked list. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: main()

Menu driven interface that invokes all linked-list functions. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: add_begin()

Allocates memory, accepts student data and inserts a node at the beginning. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: add_end()

Traverses to the last node and appends a new record. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic

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Function: add_middle()

Inserts a node in sorted order using roll number. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: print_node()

Displays all records stored in the linked list. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: count_node()

Counts and returns the total number of nodes. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: save_file()

Stores linked-list data into std.txt. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: read_file()

Reads records from std.txt and rebuilds the linked list. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: search_node()

Searches records by student name using strcmp(). Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: reverse_print()

Prints nodes in reverse order without modifying links. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: print_rec()

Uses recursion to print all nodes. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: reverse_rec()

Uses recursion to print nodes in reverse sequence. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: delete_node()

Deletes a node based on roll number. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: delete_all()

Frees all dynamically allocated nodes. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: sort_data()

Sorts records according to roll number using data swapping. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Function: reverse_link()

Reverses actual node links and updates the head pointer. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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File Handling

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Memory Management

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Advantages

Dynamic size, efficient insertion/deletion, reusable modular functions. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Limitations

Linear search complexity and extra traversal time. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Applications

Student database systems, inventory systems and record management applications. Detailed explanation: The source code implementation demonstrates the logic, algorithmic flow, pointer manipulation, traversal strategy, and operational behavior of the linked list. Understanding this function helps in mastering dynamic data structures and memory management in C.

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Conclusion

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